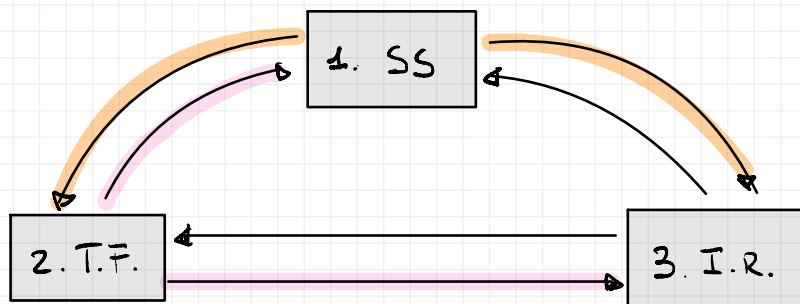




1

$$\begin{cases} x(t+1) = F x(t) + G \cdot u \\ y(t) = H x(t) + D \cdot u \end{cases}$$

2

$$y(t) = W(z) u(t)$$

↳ TRANSFER FUNCTION

$$W(z) = \frac{b_0 z^m + \dots + b_m}{z^m + \dots + a_m} \quad \text{TRANSFER FUNCTION}$$

$$= \frac{b_0 + \dots + b_m z^{-m}}{1 + \dots + a_m z^{-m}} \quad \text{FILTER FORM}$$

3

$$y(t) = \sum_{\tau=0}^{+\infty} w(\tau) u(t-\tau) = f(u(t), u(t-1), u(t-2), \dots)$$

↳ SAMPLES OF
= $w(t) * u(t)$ THE SYSTEM IMPULSE RESPONSE

SPECIAL CASES

• STRICTLY PROPER

- $D = 0$
- $B(z) < A(z)$ ORDER POLYNOMIALS
- $w(0) = 0$ (POSSIBLY ALSO OTHERS)

• FIR / IIR

	FIR	IIR
$H(z)$	ALL POLES w $z=0$ (T.F.) NO POLES (FILTER FORM)	POLES LOCATED ANYWHERE
$w(z)$	$w(i) = 0$ $i \geq m+1$	$w(z) \neq 0 \quad \forall z$

COMMENTS/NOTES

1) IT EXISTS AN ADDITIONAL REPRESENTATION 2^*

$$y(t) = f(u(t), u(t-1), \dots, \underbrace{y(t-1), y(t-2), \dots}_{\text{RECURSIVE PART}})$$

• CAN BE SIMPLY COMPUTED STARTING FROM 2

2) 3 2^* ARE MEANINGFUL FOR SYSTEM IDENTIFICATION BECAUSE:

- I/O DATA
- TIME DOMAIN

3) 1 2 ARE MEANINGFUL FOR CONTROL PURPOSES

EXERCISE # 1

$$\begin{cases} x_1(t+1) = -x_1(t) - x_2(t) + u(t) \\ x_2(t+1) = x_1(t) + x_2(t) \\ y(t) = 2x_1(t) + 3x_2(t) \end{cases}$$

S.S.

↳ STRICTLY PROPER

⇒ WRITE THE SYSTEM IN THE MATRIX FORM

$$\begin{aligned} \bar{F} &= \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}_{2 \times 2} & \bar{G} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{2 \times 1} & \bar{H} &= \begin{bmatrix} 2 & 3 \end{bmatrix}_{1 \times 2} & \bar{D} &= [0] \end{aligned}$$

⇒ COMPUTE THE SYSTEM TRANSFER FUNCTION

① * MATRIX FORMULA

$$W(z) = \bar{H} (z\bar{I} - \bar{F})^{-1} \cdot \bar{G} + \bar{D}$$

* USEFUL IF SYSTEM'S ORDER $n \leq 2$

$$\bullet (z\bar{I} - \bar{F}) = \begin{bmatrix} z+1 & 1 \\ -1 & z-1 \end{bmatrix}$$

$$\bullet (z\bar{I} - \bar{F})^{-1} = \frac{1}{\det(z\bar{I} - \bar{F})} \cdot \begin{bmatrix} z-1 & -1 \\ 1 & z+1 \end{bmatrix}$$

$$\hookrightarrow (z+1)(z-1) + 1 = z^2$$

$$\bullet W(z) = \begin{bmatrix} 2 & 3 \end{bmatrix} \frac{1}{z^2} \cdot \begin{bmatrix} z-1 & -1 \\ 1 & z+1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{z^2} \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} z-1 \\ 1 \end{bmatrix} = \boxed{\frac{2z+1}{z^2}} \quad \text{TRANSFER FUNCTION}$$

$$= \frac{z^{-2}}{z^{-2}} \cdot \frac{2z + 1}{z^2}$$

$$= \frac{2z^{-1} + z^{-2}}{1}$$

FILTER FORM

NOTE: This system is FIR, since it has all poles w $z=0$

(2) $x(t+1) \longrightarrow x(t) \cdot z$

SCHLAR WAY

z : FORWARD SHIFT OPERATOR

z^{-1} : BACKWARD SHIFT OPERATOR

$$\begin{cases} x_1 \cdot z = -x_1 - x_2 + u \\ x_2 \cdot z = x_1 + x_2 \\ y = 2x_1 + 3x_2 \end{cases} \longrightarrow x_2 = \frac{x_1}{z-1}$$

FROM FIRST EQUATION:

$$x_1(z+1) = -\frac{x_1}{z-1} + u \longrightarrow (z+1)(z-1)x_1 = -x_1 + (z-1)u$$

$$z^2 \cdot x_1 - \cancel{x_1} = -\cancel{x_1} + (z-1)u$$

$$x_1 = \frac{(z-1)}{z^2} u$$

$$x_2 = \frac{u}{z^2}$$

TRANSFER FUNCTION

$$y(t) = 2 \cdot \frac{z-1}{z^2} \cdot u + \frac{3}{z^2} \cdot u = \left[\frac{2z+1}{z^2} \right] u$$

⇒ WRITE THE RECURSIVE EQUATION 2*

- PICK THE FILTER FORM OF T.F.
- APPLY THE BACKWARD SHIFT OPERATOR

$$y(t) = \overset{W(z)}{[2z^{-1} + z^{-2}]} \cdot u(t)$$
$$= 2z^{-1} \cdot u(t) + z^{-2} \cdot u(t)$$

$$y(t) = 2 \cdot u(t-1) + u(t-2) \leadsto \text{NO RECURSIVE TERM!}$$

(DUE TO FIR SYSTEM)

⇒ WRITE THE IMPULSE RESPONSE CONVOLUTION FORM STARTING FROM S.S. REPRESENTATION

$$w(t) = \begin{cases} 0 & t=0 \text{ (FOR STRICTLY PROPER SYSTEMS)} \\ H F^{t-1} \cdot G & t>0 \end{cases}$$

REMEMBER: SINCE THE SYSTEM IS FIR, WE EXPECT TO FIND $w(3, 4, \dots) = 0$

$$w(0) = 0$$

$$w(1) = H \cdot G = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2$$

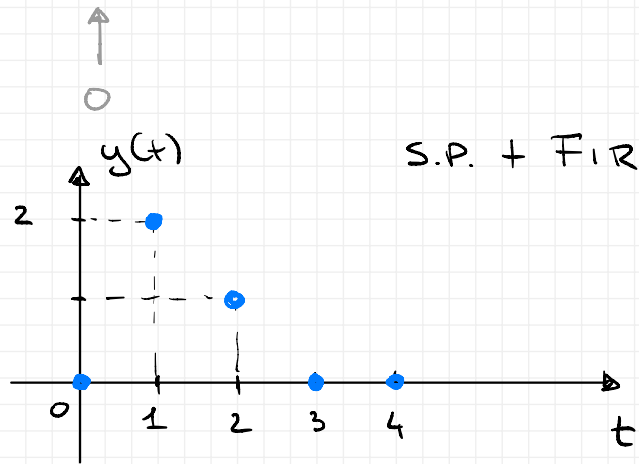
$$w(2) = H \cdot F \cdot G = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 1$$

$$w(3) = H \cdot F^2 \cdot G = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \boxed{0}$$

$$w(4) = H \cdot F^3 \cdot G = H \cdot F \cdot F^2 \cdot G = \boxed{0}$$

$$w(+\infty) = \boxed{0}$$



Exercise #2

GIVEN THE FOLLOWING SYSTEM, DESCRIBED BY THE T.F.

$$W(z) = \frac{2z^{-1} + z^{-2}}{2 - 0,6z^{-1} - 0,8z^{-2}}$$

⇒ WRITES "ONE" SYSTEM STATE SPACE REPRESENTATION

WE USE THE CONTROLLABILITY FORM / REPRESENTATION

$$W(z) = \frac{b_1 \cdot z^{n-1} + \dots + b_m}{z^n + \dots + a_n} \quad \begin{matrix} \sim "n" \text{ COEFF.} \\ \sim "n" \text{ COEFF.} \end{matrix}$$



STRICTLY PROPER SYSTEM

THEN THE S.S. CONTROLLABILITY FORM:

$$F = \begin{bmatrix} 0 & 1 & \emptyset \\ \emptyset & 0 & 1 \\ -a_n & \dots & -a_1 \end{bmatrix} \quad G = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad H = [b_m \dots b_1] \quad D = [0]$$

STEP #1 - COMPUTES THE T.F. FORM

$$\frac{2z^{-1} + z^{-2}}{2 - 0,6z^{-1} - 0,8z^{-2}} \cdot \frac{z^2}{z^2} = \frac{2z + 1}{2z^2 - 0,6z - 0,8}$$

STEP #2 - MAKES FIRST COEFF. DENOMINATOR = 1

$$\frac{0,5}{0,5} \cdot \frac{2z + 1}{2z^2 - 0,6z - 0,8} = \frac{z + 0,5}{z^2 - 0,3z - 0,4} \quad \checkmark$$

STEP #4 - APPLY THE FORMULA

$$F = \begin{bmatrix} 0 & 1 \\ 0,4 & 0,3 \end{bmatrix} \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad H = [0,5 \quad 1] \quad D = [0]$$

⇒ WRITES THE RECURSIVE FORM z^*

$$y(t) = \frac{2z^{-1} + z^{-2}}{2 - 0,6z^{-1} - 0,8z^{-2}} \cdot u(t)$$

$$y(t) = \frac{z^{-1} + 0,5 z^{-2}}{1 - 0,3 z^{-1} - 0,4 z^{-2}} \cdot u(t)$$

THE MOST CONVENIENT FORM
IS THE FILTER ONE, AND
THE FIRST COEFF. OF THE
DENOMINATOR UNITARY

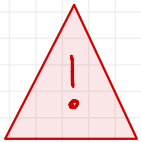
$$y(t) [1 - 0,3 z^{-1} - 0,4 z^{-2}] = [z^{-1} + 0,5 z^{-2}] \cdot u(t)$$

|

$$y(t) = 0,3 y(t-1) + 0,4 y(t-2) + u(t-1) + 0,5 u(t-2)$$

RECURSIVE TERM
(SYSTEM IS NOT FIR)

⇒ WRITE THE SYSTEM IN THE IMPULSE RESPONSE CONVOLUTION FORM



THE SYSTEM IS NOT FIR → COMPUTES ∞ NUMBER OF
SAMPLES!

LET'S DO IT ANYWAY...

$$\text{IDSA: } y(t) = \sum w(\tau) \cdot u(t-\tau)$$

$$\begin{matrix} \text{s.p.} \\ w(0)=0 \end{matrix} \rightarrow \begin{matrix} | \\ = w(1) \cdot u(t-1) + \dots + w(k) \cdot u(t-k) + \dots \end{matrix}$$

$$y(t) = \underbrace{[w(1) \cdot z^{-1} + \dots + w(k) \cdot z^{-k} + \dots]}_{W(z)} \cdot u(t)$$

WE USED TO WRITE $W(z)$ AS A SUM OF NEGATIVE POWERS OF z .
THE COEFF. WILL BE THE IMPULSE RESPONSE WE ARE LOOKING FOR



LONG DIVISION METHOD

$$W(z) = \frac{z^{-1} + 0,5 z^{-2}}{1 - 0,3 z^{-1} - 0,4 z^{-2}}$$

SUGGESTED
REPRESENTATION

$$z^{-1} + 0,5z^{-2}$$

$$z^{-1} - 0,3z^{-2} - 0,4z^{-3}$$

$$0,8z^{-2} + 0,4z^{-3}$$

$$0,8z^{-2} - 0,24z^{-3} - 0,32z^{-4}$$

$$0,64z^{-3} + 0,32z^{-4}$$

$$0,64z^{-3} - 0,192z^{-4} - 0,256z^{-5}$$

$$0,512z^{-4} + 0,256z^{-5}$$

$$1 - 0,3z^{-1} - 0,4z^{-2}$$

$$z^{-1} + 0,8z^{-2} + 0,64z^{-3} + \dots$$

$$\hookrightarrow W(z) = \frac{z^{-1} + 0,8z^{-2} + 0,64z^{-3}}{1 - 0,3z^{-1} - 0,4z^{-2}} + \frac{0,512z^{-4} + 0,256z^{-5}}{1 - 0,3z^{-1} - 0,4z^{-2}}$$

$$w(0) = 0$$

$$w(1) = 1$$

$$w(2) = 0,8 \Rightarrow w(t) = (0,8)^{t-1}$$

$$w(3) = 0,64$$

⋮

EXTRA: LONG DIVISION USING ANOTHER FORM OF THE T.F.

LET'S CONSIDER, FOR INSTANCE:

$$W(z) = \frac{2z + 1}{2z^2 - 0,6z - 0,8}$$

APPLYING THIS LONG DIVISION, WE HAVE:

$\begin{array}{r} 2z + 1 \\ 2z - 0,6 - 0,8z^{-1} \\ \hline 1,6 + 0,8z^{-1} \\ 1,6 - 0,48z^{-1} - 0,64z^{-2} \\ \hline 1,28z^{-1} + 0,64z^{-2} \\ 1,28z^{-1} - 0,384z^{-2} - 0,512z^{-3} \\ \hline 1,024z^{-2} + 0,512z^{-3} \end{array}$	$\begin{array}{r} 2z^2 - 0,6z - 0,8 \\ \hline z^{-1} + 0,8z^{-2} + 0,64z^{-3} + \dots \end{array}$
--	--

SO WE HAVE FOUND THAT:

$$W(z) = \left[z^{-1} + 0,8z^{-2} + 0,64z^{-3} \right] + \frac{1,024z^{-2} + 0,512z^{-3}}{2z^2 - 0,6z - 0,8}$$

$$\begin{array}{l} \downarrow \\ \left. \begin{array}{l} w(0) = 0 \\ w(1) = 1 \\ w(2) = 0,8 \\ w(3) = 0,64 \end{array} \right\} \text{SAME RESULT AS BEFORE!} \end{array}$$